

Kubernetes & Google Kubernetes Engine (GKE)

by Akash Agrawal



Agenda

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- Understanding Kubernetes
 - Traditional & Containerized App Deployment
 - Understanding Containers
 - Understanding Kubernetes
 - Kubernetes Fundamentals
 - Kubernetes Resources/Objects
 - Kubernetes Flow Diagram
 - Kubernetes Cluster Architecture

Agenda

- Understanding Google Kubernetes Engine (GKE)
 - Understanding GKE
 - GKE Features

About Me

About Me

- Google Developer Expert (on **Google Cloud Platform** Category)
- 11 years of experience in IT Industry
- Worked with various clients like Sabre/Citi Bank/Goldman Sachs/L&T Infotech etc.
- Currently I work as Independent Consultant (as Technical Adviser/Architect/Trainer) & Tech Evangelist

Understanding Kubernetes

Traditional & Containerized App Deployment

- Traditional App Deployment
- Problems with Traditional App Deployment
- Containers & what problems does it solve
- Kubernetes (Container Orchestrator) & what problems does it solve

Understanding Containers

- Package your application/component & it's dependencies
- Container Runtimes : Docker / Rkt etc.
- Focused around : Build / Ship / Run
 - Build: build container on local laptop manually or automate it through CI/CD Pipeline
 - Ship: ship through Container Registry e.g. Docker Hub/GCR etc.
 - Run: Run anywhere e.g. Cloud/On-Prem etc.

Understanding Kubernetes

- Kubernetes is an Open Source system for Container Orchestration (Deployment/Management/Scaling)
- Focused around:
 - Scheduling
 - Scaling (Manual / Auto)
 - Self Healing or Auto Repairing
 - ...

Kubernetes Fundamentals

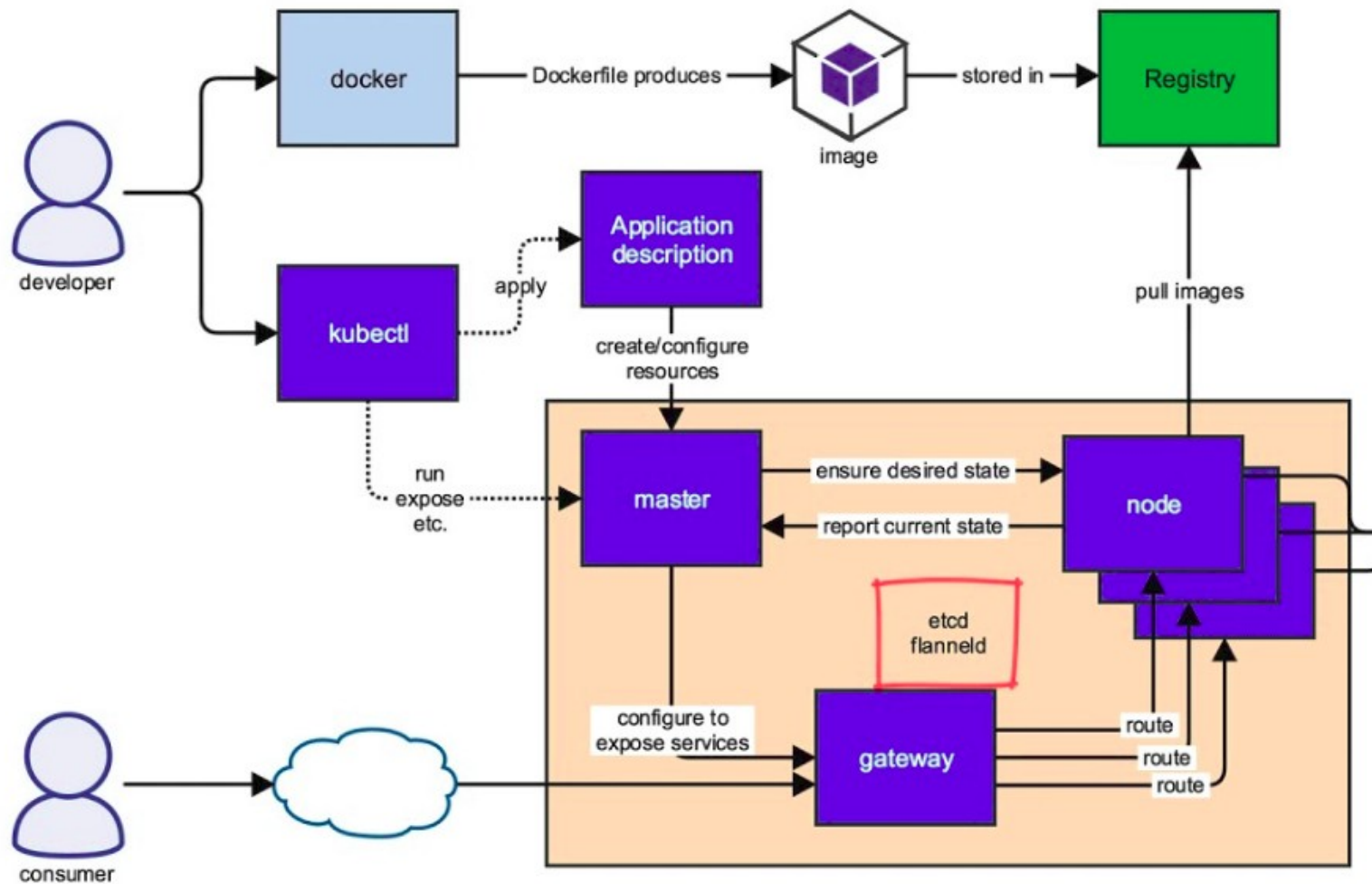
- Kubernetes Cluster
- Kubernetes Artifacts/Resources/Objects
- Containerized Applications

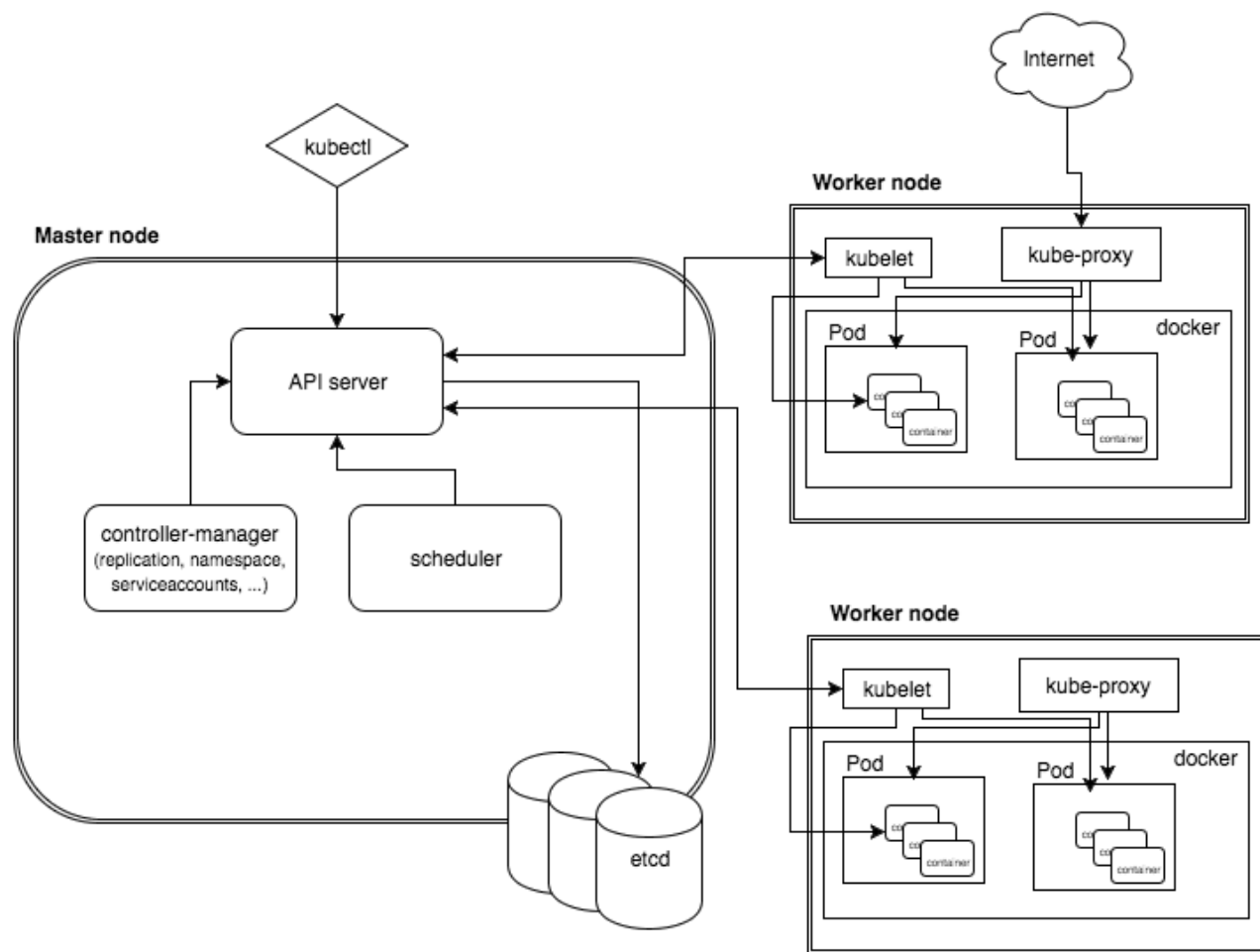
Kubernetes Resources/Objects

- Pod (Single/Multi Container Pod)
- Service (ClusterIP/NodePort/LoadBalancer)
- Controller Objects:
 - ReplicaSet
 - Deployment
 - StatefulSet
 - Job
 - ...

Kubernetes Resources/Objects

- Namespace
- ConfigMap
- Secret
- Label & Selectors





Kubernetes Cluster Architecture

- Control Plane

- Focused around managing Cluster & its State
- Components:
 - Kube-apiserver
 - Etcd
 - Kube-controller-manager
 - Kube-scheduler

Kubernetes Cluster Architecture

● Data Plane

- Focused around managing individual Nodes & deployed Workloads
- Components:
 - Kubelet
 - Kube-proxy
 - Container Runtime

Understanding Google Kubernetes Engine (GKE)

Understanding GKE

- Managed Solution for Kubernetes on Google Cloud
- Provides Reliable, Efficient & Secured way to run Kubernetes Clusters (on GCP)
- Focused around:
 - Cluster Scaling
 - Self Healing or Auto Repairing of Nodes
 - Node Pools : CPU/GPU Pools
 - ...

GKE Features

● Features:

- Fully Managed
- Auto Scaling / Auto Upgrade / Auto Repair
- Integration : IAM / StackDriver / VPC
- Security, Compliance, Runs on Optimized OS (COS)
- Accelerators Support : GPUs/TPUs
- Various Cluster Topologies : Zonal Clusters / Regional Clusters
- Workload Portability : On-Premises / Cloud
- ...

Look for:

- Qwik Labs:

- Kubernetes in Google Cloud: <https://www.qwiklabs.com/quests/29>
- Kubernetes Solutions: <https://www.qwiklabs.com/quests/45>

Google Cloud Platform - Resources

- Google Cloud Platform 101 (Cloud Next '19):
<https://www.youtube.com/watch?v=vmOMataJZWw>
- Google Cloud Developer Cheat Sheet:
<https://raw.githubusercontent.com/gregsramblings/google-cloud-4-words/master/Poster-medres.png>
- 100+ announcements from Google Cloud Next '19:
<https://cloud.google.com/blog/topics/inside-google-cloud/100-plus-announcements-from-google-cloud-next19>

Google Cloud Platform - Resources

- Google Cloud Next '19 Sessions:

<https://www.youtube.com/playlist?list=PLlivdWyY5sqIXvUGVrFuZibCUdKVzEoUw>

- GCP Certification Resources:

<https://github.com/ddneves/awesome-gcp-certifications>

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Thanks